# JOSUÉ LEIVA POBLETE

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## SUMMARY

Third-year Telematics Engineering student (UTFSM), research assistant at LabOPT and teaching assistant in programming and calculus. I build and operate real systems: a C++ edge runtime for IoT gateways, a web platform in production and a 34-container home server, with hands-on networking (VLANs, VLSM, routing, firewalls). Seeking a 320-hour Industrial Internship, available Dec 5, 2026 – Mar 15, 2027, ideally in software development, telecommunications, data centers or IoT.

## EDUCATION

**Universidad Técnica Federico Santa María (UTFSM)** *Expected December 2029*
B.S. in Civil Telematics Engineering (3rd year, on track) *Reconocimiento al Mérito Académico, 2025 and 2026*

## EXPERIENCE

**LabOPT – Optimization and Applied Computing Research Lab** Mar 2026 – Nov 2026
*Research Assistant – Edge Runtime Developer (Plazas Inteligentes)* *Santiago, Chile*

- Develop the C++17 runtime for Raspberry Pi 5 gateways of an urban telemetry network (3 gateways, each with 3 sensors, an ADC and a still-image camera), in a team of three.
- Designed a multithreaded pipeline with thread-safe queues so capture never blocks on sensors or uploads; queues buffer data when the uplink is down. Found and fixed a mutex self-deadlock in queue access.
- Implemented HTTP multipart uploads, remote control via Server-Sent Events, link-quality reporting, hardened I2C drivers, automatic restarts and graceful shutdown.

**Universidad Técnica Federico Santa María** Aug 2025 – Jun 2026
*Teaching Assistant – Programming Seminar (TEL20125), Calculus I (MAT071)* *Santiago, Chile*

- Programming Seminar: taught C and C++ in lecture-style sessions and graded student work. Calculus I: graded exams and taught Wolfram Mathematica in labs.

**freeCodeCamp – Spanish Translation Team** Feb 2022 – Oct 2024
*Volunteer Translator* *Remote*

- Translated programming tutorials and articles from English to Spanish.

## PROJECTS

**Web SOS – Evangelism Management Platform** In production since 2026
*Full-Stack Developer and Sysadmin (volunteer)* *Python, Django, HTMX, PostgreSQL, Docker, Nginx*

- Built and deployed a containerized platform for a faith-based community, replacing paper and chat records with a central system for contacts, follow-ups and approved publishing; designed the PostgreSQL schema with role hierarchies.

**UBike IoT System – Smart Bike Parking** Academic project
*Developer* *ESP32 (C++), RFID, Django, Wi-Fi*

- Built ESP32 locking modules that authenticate RFID keychains over Wi-Fi against a central Django server, with a lockdown mode for tampering or power loss.

## TECHNICAL SKILLS

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| **Networking** | Subnetting, VLSM, VLANs, routing, firewalls, Wireshark, Cisco Packet Tracer, Huawei equipment, HTTP/SSE, TCP/UDP sockets |
| **Infrastructure** | Linux server admin (34 containers, Docker, Nginx, Cloudflare Tunnels, Beszel monitoring, RAID 1, backups), Git |
| **Programming** | C++ (5-star HackerRank), Python, SQL, C, Java; Django, REST APIs, PostgreSQL |
| **Coursework** | Computer Networks Lab, Databases, Software Design, Cryptography (in progress) |
| **Languages** | Spanish (native), English (C1, Busuu certificate) |